1. In the 4 ethical theories talked about in class, which of the following best describes Mills Utilitarianism?
a) You have a fundamental duty to act in an ethical manner
b) You must achieve a "golden mean" in balancing ethical dilemmas
c) Best course of action is a solution that produces the maximum benefit for the greatest number of people
d) (b) and (c) above
e) All of the above
Answer: c

2. In the 4 ethical theories talked about in class, which of the following best describes Kant's Formalism?
a) You have a fundamental duty to act in an ethical manner
b) You must achieve a "golden mean" in balancing ethical dilemmas
c) Best course of action is a solution that produces the maximum benefit for the greatest number of people
d) You have a fundamental right to life, dignity, liberty
e) PEO code of ethics
Answer: a

3. In the 4 ethical theories talked about in class, which of the following best describes Locke's Rights Ethics?
a) You have a fundamental duty to act in an ethical manner
b) You must achieve a "golden mean" in balancing ethical dilemmas
c) Best course of action is a solution that produces the maximum benefit for the greatest number of people
d) You have a fundamental right to life, dignity, liberty
e) PEO code of ethics
Answer: d

4. In the 4 ethical theories talked about in class, which of the following best describes Aristotle's Virtue Ethics?
a) You have a fundamental duty to act in an ethical manner
b) You must achieve a "golden mean" in balancing ethical dilemmas
c) Best course of action is a solution that produces the maximum benefit for the greatest number of people
d) You have a fundamental right to life, dignity, liberty
e) PEO code of ethics
Answer: b

5. Metrolinx is building the Union-Pearson rail link to bring people from the airport to Union Station via a dedicated direct powered train. Although this will expose people living in the Weston and Junction areas of Toronto to more pollution, using tax money to build this line is an example of

a) Mills Utilitarianism
b) Kant's Formalism
c) Locke's Rights Ethics
d) Aristotle's Virtue Ethics
e) None of the above
Answer: a

6. The Canadian Charter of Rights and Freedoms embodies many of the same concepts of this ethical theory
a) Mills Utilitarianism
b) Kant's Formalism
c) Locke's Rights Ethics
d) Aristotle's Virtue Ethics
e) None of the above
Answer: c

7. In deciding on the design of the new Ryerson Student Learning Center being built on Gould and Yonge Street, the Ryerson building committee most likely applied this ethical theory
a) Mills Utilitarianism
b) Kant's Formalism
c) Locke's Rights Ethics
d) Aristotle's Virtue Ethics
e) None of the above
Answer: d

8. Ethical problems relating to engineering activities in companies that manufacture compressed gas cylinders are likely to arise during
a) The selection of a supplier of pressure relief valves, where the engineer making the decision does so in an unbiased and reasonable manner
b) The selection of a supplier of pressure relief valves, where the engineer making the decision does so based on reasonable engineering factors but where the choice is likely to be perceived as biased
c) The selection of a supplier of pressure relief valves, where the engineer making the decision does so while biased by an interest that can benefit a relative
d) The selection of a supplier of pressure relief valves, where the engineer making the decision does so following precisely the approved procedures and criteria stipulated by the engineer's company
e) (b) and (c)
Answer: e

9. Consider the code of ethics of the engineering society IEEE and the provincial association PEO
a) Both focus on licensing in the electrical and electronics engineering field
b) Both focus on licensing in the province of Ontario
c) There are some similarities in the two codes of ethics

d) There are no differences in the two codes of ethics
e) (b) and (c)
Answer: c

10. An engineer who takes on an administrative management position for her company
a) Is permitted to participate in and oversee engineering activities
b) Is permitted to participate in and oversee design activities
c) Is permitted to participate in and oversee non-engineering activities
d) (a) and (b)
e) All of the above
Answer: e

11. Of the duties of the engineer alluded to in PEO's engineering code of ethics, which is considered to be of paramount importance?
a) Duty to subordinates who work under your guidance
b) Duty to society
c) Duty to self
d) Duty to suppliers of raw materials used in your company
e) Duty to your manager
Answer: b

12. An engineer finds himself faced with a serious ethical dilemma in an engineering matter within a branch office of his company. Which of the following factors should he take into account in deciding the course of action to follow?
a) The impact his decision can have on his company's engineering and non-engineering staff members
b) The impact his decision can have on only his company's engineering staff members
c) The impact his decision can have on only his company's engineering staff members who are employed at the branch office
d) (b) and (c)
e) All of the above
Answer: a

13. An engineer who quits a company in order to establish and run her own engineering company is likely to find herself facing
a) Fewer ethical dilemmas than when she was working at her past company as an engineer
b) More ethical dilemmas than when she was working at her past company as an engineer
c) Fewer ethical dilemmas than other engineers who work at her past company as engineers
d) More ethical dilemmas than other engineers who work at her past company as engineers
e) (b) and (d)
Answer: e

14. A mechanical product is sold by an engineering company to consumers through hardware stores. The mechanical product is found to have a structural problem that causes 46 serious injuries in the first six months after it goes on sale to the public, based on a report by the

company's safety engineers. A prudent course of action would be to
a) Recall the products that have been sold so that repairs can be made to correct the structural problem
b) Recall the products that have been sold so that repairs can be made to correct the structural problem, and offer the owners of products that have been sold financial compensation
c) Correct the structural problem so that the reason for the mechanical product causing injuries is eradicated
d) (a) and (b)
e) All of the above
Answer: e

15. Consider two licensed engineers in Ontario, who work for different companies. The duty of Engineer A to Engineer B
a) Exceeds Engineer A's duty to another engineer who works for Engineer A's company
b) Is below the engineer's duty to another engineer who works for Engineer A's company
c) Exceeds Engineer A's duty to protect the public safety
d) Is below Engineer A's duty to protect the public safety
e) (b) and (d)
Answer: d

16. An experienced licensed engineer has had an engineering seal for more than two decades. A charge of professional misconduct could reasonably be made if the licensed engineer
a) Applies the engineering seal to an engineering drawing without reviewing it
b) Applies the engineering seal to an engineering drawing that is complete and properly done
c) Applies the engineering seal to an engineering drawing that is complete, except it is lacking one part
d) (a) and (c)
e) All of the above
Answer: d

17. An engineer is an engineering manager in a health product company, which relies in part on public donations for part of its funding. As a consequence of the code of ethics in the province where the engineer works, the engineer
a) Can dispute orders made by the president of the company
b) Can dispute orders made by the president of the company, but only when they endanger the public welfare
c) Can dispute requests made by donors who provide money to the company through donations
d) (b) and (c)
e) All of the above
Answer: d

18. There is evidence that climate change will result in
a) An increased frequency and intensity of storms
b) Greater droughts and floods to exacerbate food production and distribution problems,

increasing hunger
c) Increase in hot days, causing an increase in heat-stress-related health problems, especially for children and the elderly, the poor, and those already sick
d) Deaths from natural disaster, especially in developing countries
e) All of the above
Answer: e

19. Which of the following statements are true?
a) Canada's GHG emissions are higher than our Kyoto commitment to be 6% below 1998 levels
b) It is not ethical to give or accept bribes even if it is the only way you can continue to operate in a country
c) Stakeholder engagement is good for an organization
d) An engineer should know what the rights and risks of harm to all of his or her stakeholders are
e) All of the above
Answer: e

20. Organizational governance is the system by which an organization makes and implements decisions in pursuit of its objectives
a) True
b) False
Answer: a

Scenario for Questions 21 to 25
In an effort to become better known in the engineering community, Marketwise Pipe Manufacturing Company hires Susan, an expert in marketing. Susan suggests that Marketwise offer a complimentary seminar to educate engineers on the current advances in the selection and use of pipe construction. She suggests that Marketwise extend invitations to several local engineers and that they offer refreshments at the seminar, including a buffet luncheon during the seminar and a cocktail reception immediately following.

Suppose that Marketwise takes Susan's suggestion and extends an invitation to Terry, a local engineer who is new to the area. Terry is not sure how to respond to this invitation. Being new to the area, attending such a seminar will give him the opportunity to meet and network with other engineers. Furthermore, learning more about the selection and use of pipes would be helpful to Terry. On the other hand, Terry knows he will be purchasing pipes at some point in the future and wonders if accepting the invitation might later create a conflict of interest. In addition, Terry wonders if accepting the invitation will amount to a bribe.

21. If you were an executive at Marketwise, how would you respond to Susan's suggestion?
a) Not be concerned at all
b) Be concerned that providing a free seminar and associated complimentary activities could be construed as obligating attendees to buy products from Marketwise Pipe Manufacturing Company

c) Be concerned that simply offering a free seminar intended to educate potential clients about current advances in piping is potentially unethical
d) (b) and (c)
e) None of the above
Answer: d

22. How should Terry respond to the invitation?
a) Be concerned that simply attending a seminar to learn about advances in piping is potentially unethical
b) Ensure that attending the free seminar and associated complimentary activities in no way obligates him to buy products from Marketwise Pipe Manufacturing Company now or in the future
c) Not be concerned at all
d) (a) and (b)
e) None of the above
Answer: b

23. Terry's concern about a potential conflict of interest is
a) Not relevant because he is new to the area
b) Only relevant because if he has a job presently, since he can have no conflict of interest if he is unemployed
c) Only relevant if he agrees to give preference to Pipe Manufacturing Company because of attending
d) (b) and (c)
e) None of the above
Answer: c

24. The potential conflict of interest mentioned in the case study is attributable to the fact that
a) Engineers should know about advances in piping without attending the type of seminar offered by Marketwise Pipe Manufacturing Company
b) Marketwise Pipe Manufacturing Company is offering complimentary attendance only to a limited number of local engineers
c) The seminar will provide opportunities for attendees to network with other engineers
d) All of the above
e) None of the above
Answer: e

25. In the future, Terry actually purchases pipes from Marketwise Pipe Manufacturing Company
a) He is not in violation of the PEO Code of Ethics if he did so because it was the most beneficial option based on his engineering judgment
b) He is in violation of the PEO Code of Ethics, even if he did so because it was the most beneficial option in his engineering judgment, because he learned about advances in piping at the seminar
c) He is clearly in violation of the PEO Code of Ethics
d) (b) and (c)

e) None of the above
Answer: a

Scenario for Questions 26 to 30
A provincial agency issues a request for proposals for the design and construction of a new highway overpass in a rapidly growing urban area. Because the overpass will carry a large volume of daily traffic, including public transit vehicles and emergency services, the project is regarded as one with important public safety implications. The agency invites several engineering firms to submit bids and states in its proposal documents that cost will be a significant factor in the evaluation, but not necessarily the deciding factor. It also states that all proposals must satisfy the required engineering, safety, and performance standards.

Three firms are selected as finalists: the Allied Corporation, Mikie Incorporated, and the O'Leary Company. After reviewing the submissions, the provincial agency awards the contract to the O'Leary Company. The O'Leary proposal is the lowest-cost bid of the three, but the agency explains that the proposal was accepted because, in its judgment, it met the technical requirements, addressed public safety appropriately, and represented the best overall value.

Some principals at the Allied Corporation and Mikie Incorporated are troubled by the result. They believe that a significantly lower bid on a project of this kind could suggest that corners may have been cut or that the design may not provide the same margin of safety or quality as the other proposals. Others point out that the provincial agency had clearly stated from the beginning that cost would be an important consideration, and that any firm submitting a proposal had an obligation to remain competitive while still meeting the required standards.

Suppose that a principal at the Allied Corporation is especially concerned that the public could be put at risk if the winning firm underestimated the true cost of completing the project safely. Suppose also that, after the decision is announced, the Allied Corporation and Mikie Incorporated file a protest and call for a public hearing. At the hearing, questions are raised about whether the O'Leary Company was sufficiently qualified, whether the provincial agency acted consistently with its stated evaluation criteria, and whether a low-cost proposal for this type of public project can still meet proper engineering and safety standards.

26. If you are a principal in the Allied Corporation, you should
a) Express your concerns to all stakeholders involved in the proposal process, with the request that assurances be made that the safety of the public will not be jeopardized
b) Quit your job and try to get one with the O'Leary Company
c) Launch a lawsuit against Mikie Incorporated for putting forward such a high-cost proposal
d) All of the above
e) None of the above
Answer: a

27. If, following the decision, the Allied Corporation and Mikie Incorporated file a protest with the agency and call for a public hearing based on the ideas expressed above, how would you respond if you were a principal with the O'Leary Company?

a) Explain how the qualifications of your company were vetted by the provincial agency and deemed acceptable for inclusion in the three finalists
b) Explain that the costs of your company's bid were kept as low as reasonably possible because the provincial agency stated cost would be a significant factor
c) Explain how the project can be done safely and to proper engineering standards for the cost your company put forward in its bid
d) (a) and (c)
e) All of the above
Answer: e

28. Should the provincial agency have accepted this bid after maintaining that cost would not be the overriding factor?
a) Yes, if all relevant factors were considered carefully and fairly in picking the winning bid
b) Yes, if the bid was the best in overall terms
c) No, because the bid had the lowest cost
d) No, even if the bid was the best in overall terms because it nonetheless had the lowest cost
e) (a) and (b)
Answer: e

29. The stipulation by the provincial agency that cost will be a significant factor in evaluating bids but not necessarily the deciding factor
a) Is unethical because it encourages cutting corners in designs to achieve the lowest bid
b) Is ethical because it encourages containing costs in designs to achieve a low bid while still meeting the design objectives
c) Means all the bidding companies can win the competition if they can achieve the lowest cost bid
d) Puts the safety of the public at risk
e) All of the above
Answer: e

30. What information would be relevant to your ethics assessment of this case?
a) The engineering competency of the bidding firms
b) The salaries paid to the presidents of the bidding firms over the last year
c) How public safety is addressed in the proposals
d) (a) and (c)
e) All of the above
Answer: d

Scenario for Questions 31 to 35
Engineer Alpha is employed by a large consulting engineering firm and is responsible for reviewing and recommending components for use in a municipal water distribution project. As part of his duties, Alpha evaluates several types of industrial valves offered by different manufacturers. After considering the technical specifications, performance history, and project requirements, Alpha recommends that a particular manufacturer’s valves be specified for the project, and a large number of these valves are subsequently purchased.

Some time after the valves have been specified and purchased, a salesperson from the valve manufacturer visits Alpha. During the visit, the salesperson thanks Alpha for his company's business and offers Alpha an all-expenses-paid trip to a resort outside the country. The trip would include airfare, hotel accommodation, meals, and entertainment. The salesperson explains that the trip is intended as a gesture of appreciation for Alpha's support of the manufacturer's products and says that several other engineers from different firms have also been invited.

Alpha realizes that he made his recommendation before knowing anything about the proposed trip. Nevertheless, he wonders whether accepting such a generous gift could create an ethical problem. He considers whether accepting the invitation might affect, or appear to affect, his independent professional judgment in future decisions involving the same manufacturer. He also considers whether members of the public, his employer, or other suppliers might reasonably view his acceptance of the trip as improper or as placing him in a conflict of interest.

Suppose that Alpha accepts the trip and later continues to participate in engineering decisions involving similar valve products from the same manufacturer.

31. Engineer Alpha can
a) Accept the trip, since he selected the valves without knowledge of this invitation
b) Contact his supervisor and ask for permission in accepting the trip
c) Not accept such a big gift
d) Be perceived publicly as not behaving ethically if he accepts the gift
e) Both (c) and (d)
Answer: e

32. If Engineer Alpha accepts the trip
a) His behavior is unlawful and disgraceful
b) He is guilty of professional misconduct
c) He is endangering the public
d) He may be in breach of the Code of Ethics
e) None of the above
Answer: d

33. If Engineer Alpha is taking the trip, he is
a) Acting as a faithful agent for his employer
b) Acting in the public interest
c) Acting in his own interest
d) In a situation that may be perceived as a conflict of interest
e) Both (c) and (d)
Answer: e

34. In judging the above case, which Code of Ethics paragraphs apply?
a) 4
b) 7.i

c) 7.iv
d) (a) and (c)
e) 8
Answer: d

35. If the last paragraph of the above case study read as follows: after a large number of valves had been specified and purchased, the salesperson of the valve manufacturer visited Alpha and invited him to participate in a seminar describing new improved products. The seminar would take place at a hotel in town and would be followed by a cocktail reception and dinner. Alpha attends the seminar. His behavior is
a) Unethical, because by attending the free seminar he will recommend the product
b) Not a compromise of the engineer's independent professional judgment
c) Ethical, because the cocktail reception and dinner are a courtesy provided to engineers in the area
d) An opportunity to become educated on a particular topic
e) (b), (c), and (d)
Answer: e

Scenario for Questions 36 to 40
Omega is a professional engineer employed by ABC Manufacturing Company, a firm that produces specialized metal components for industrial equipment. ABC has entered into a contract with Good Engineering Company to manufacture and supply a large shipment of precision parts that will eventually be used by XYZ in a system where reliability and safety are important. The contract documents and technical specifications clearly state the type and grade of metal alloy that must be used in fabricating the parts.

During production, Sigma, Omega's supervisor at ABC, becomes concerned that the specified alloy is too expensive and will reduce the company's profit on the contract. Sigma instructs production staff to substitute a cheaper alloy that is similar in appearance but does not meet the agreed specifications. Sigma tells Omega that the substitute material should still be "good enough" for the intended purpose and says that the change is necessary if ABC is to remain competitive and profitable. Omega is troubled by this instruction because he knows that the parts no longer meet the contract requirements and that the customer has not approved any such substitution.

Omega raises concerns internally, pointing out that the use of a cheaper alloy may affect performance, may mislead the customer, and may expose both the company and the engineers involved to serious ethical and legal consequences. Sigma dismisses Omega's concerns and insists that the shipment proceed as planned. As the shipment is being prepared for delivery to Good Engineering Company, Omega is asked to sign a report certifying that the parts comply with the contractual specifications.

Omega now must decide whether he can ethically sign the report, whether refusing to do so is enough, and what responsibilities he has in light of the company's decision to proceed with the cheaper parts.

36. Omega has an ethical dilemma relating to the following paragraphs of the Code of Ethics
a) 1.i
b) 1.iii
c) 3
d) 7.i
e) All of the above
Answer: e

37. Omega is not pursuing the issue any further and ABC is producing a cheaper part. As the shipment is prepared to be sent to XYZ, Omega is asked to sign a report verifying that the specifications for the part have been met. Which one of the following actions does **not** violate the Code of Ethics?
a) Omega decides that he will not sign the report, knowing that a cheaper part has been produced
b) Omega signs the report at the instruction of his boss Sigma
c) Omega signs the report and places an anonymous phone call to Good Engineering Company informing them about the changes
d) Omega takes sick leave in order not to sign the report
e) None of the above
Answer: a

38. Omega refuses to sign the report. However, Sigma persuades another engineer, Theta, to sign the report. The cheaper parts are shipped to Good Engineering Company.
a) ABC is committing fraud
b) Both engineers Sigma and Theta are acting unethically
c) Both engineers Sigma and Theta are breaking the law
d) By acting in the interest of increasing profit illegally, ABC is jeopardizing its future
e) All of the above
Answer: e

39. In judging the above case, which ethical theory would justify Omega's decision not to sign the report?
a) Duty ethics
b) Rights ethics
c) Utilitarian ethics
d) Virtue ethics
e) All of the above
Answer: e

40. Replacing the metal used in fabrication of the parts with a cheaper alloy means that both ABC and its engineers
a) May be vicariously liable
b) Are committing fraud
c) Are breaking the contract
d) Are behaving unethically

e) All of the above
Answer: e

Scenario for Questions 41 to 46
A developer proposes the construction of a small commercial building in Ontario and obtains a complete set of design drawings from a foreign engineering firm. The drawing package includes structural, architectural, plumbing, electrical, and mechanical drawings. Because the building is to be constructed in Ontario, the developer asks BCEC, a local consulting firm specializing in structural engineering, to review the drawings and provide the necessary professional engineering seal for submission to the local authority having jurisdiction.

Engineer Kappa, a principal at BCEC, is experienced in structural engineering and is asked to check the structural loadings and the integrity of the structural design. Kappa carefully reviews the structural drawings and confirms that the structural aspects of the design are satisfactory and conform to the Ontario Building Code. However, Kappa does not carry out a detailed review of the plumbing, electrical, or other non-structural drawings, since these areas are outside BCEC's retained scope and outside his area of specialization.

Despite this limited review, Kappa places his professional engineer's seal on the entire package of drawings without clearly indicating that his review was restricted to structural matters only. The complete sealed package is then prepared for submission as though all drawings had been checked and accepted under Kappa's professional responsibility. Later, concerns are raised that some of the non-structural drawings may not conform to Ontario requirements, even though the structural drawings themselves are correct.

Suppose that the building is ultimately never constructed. Questions are then raised about whether Kappa acted properly in sealing the entire drawing package, what his responsibilities were when applying his seal, and whether it matters that the original drawings were prepared by a foreign firm rather than an Ontario-based one.

41. When engineer Kappa checked the drawings for structural loadings
a) He was correct to stamp all drawings
b) He was supposed to check conformity to the Ontario Building Code
c) He was responsible for all the information contained in the drawings
d) Both (b) and (c)
Answer: d

42. The proposed building was never built
a) This fact has no bearing when evaluating engineer Kappa's behavior
b) If the building was not built, nobody is at fault
c) Omicron was not qualified to check the plans
d) None of the above
Answer: a

43. Engineer Kappa and his company BCEC were specialized in structural engineering and had been paid to check the integrity of the structure. The structural drawings were correct and not in violation of the Ontario Building Code. The problems were with the other drawings. Using the professional engineer stamp on all drawings was in violation of the Code of Ethics, especially
a) 2.iii
b) 1.v
c) 4
d) 3
e) 7.iii
Answer: b

44. What if engineer Kappa were diligent and had stamped only the structural drawings and had made a clear statement on all other drawings, such as plumbing and electrical, that only the structural checking was completed by him?
a) Would not have made any difference in judging the case
b) Would satisfy that he was doing his duty to the best of his knowledge in structural engineering
c) Would be judged as ethical behavior
d) Both (b) and (c)
e) None of the above
Answer: d

45. If the building had been built and the drawings, stamped by engineer Kappa, submitted in construction
a) Public safety might have been endangered
b) The owner of the building would have been in violation of the Ontario Building Code
c) Liability in law might have occurred
d) All of the above
e) None of the above
Answer: d

46. The drawings were completed by a foreign firm, for example from the USA
a) The laws in Canada prohibit the use of foreign firms, so the project must be done by a Canadian firm
b) The drawings must conform with the rules and regulations in the jurisdiction for which they are intended to be used
c) Engineer Kappa was not allowed to seal drawings done by a foreign firm
d) A professional engineer's seal on a drawing implies that the person applying it has checked the drawing for conformity with local statutes and regulations
e) Both (b) and (d)
Answer: e

47. Behavior that is in accordance with accepted principles of right or good conduct in the context of a particular situation, and consistent with international norms of behavior, is considered ethical

a) True
b) False
Answer: a

48. Canadian organizations do not need a code of conduct that deals with human rights issues because human rights abuses do not happen in Canada
a) True
b) False
Answer: b

49. The ability to influence comes with the responsibility to exercise that influence
a) True
b) False
Answer: a

50. You are accountable for harm caused by your decisions, even if you did not know your decisions would cause harm
a) True
b) False
Answer: a

51. The most common complaint about globalization is that it widens the gap between the rich and the poor.
a) True
b) False
Answer: a

52. As a licensed engineer in Ontario, your duty under the province's engineering code of ethics to another engineer who works for a multinational company that supplies components to your company is:
a) Equal to your duty to protecting the public welfare
b) Less than your duty to protecting the public welfare
c) Greater than your duty to protecting the public welfare
d) (a) and (b)
e) None of the above
Answer: b

53. As a result of an engineer's duty to himself or herself, an engineer who works for a company as an employee:
a) Is permitted to dispute the company's requests
b) Should keep up to date with the relevant emerging engineering codes
c) Should uphold the intent of the engineering code of ethics
d) (b) and (c)
e) All of the above
Answer: d

54. Ethical dilemmas relating to engineering activities in telecommunications equipment production can arise during the following:
a) Choosing a supplier of electronic circuitry
b) Exploiting newly developed intellectual property in the form of an invention
c) Establishing a social club within the company
d) (a) and (b)
e) (b) and (c)
Answer: d

55. From an engineering perspective, the following activities are usually inappropriate:
a) Making a specific bid selection because it will benefit the career of your spouse, who is not an engineer and is employed with another company
b) Actions that are intended to improve your company's internal processes and procedures
c) Accepting an expensive gift to make a certain decision
d) (a) and (c)
e) All of the above
Answer: d

56. An engineer would be involved in professional misconduct if he or she:
a) Disciplines an engineer who reports to him or her without investigating the situation
b) Makes a decision when asked to do so regarding the sizing of a chemical reaction vessel in a petrochemical plant
c) Applies his or her engineering seal to an engineering circuit diagram for a combined media player and cell phone
d) (a) and (b)
e) (a) and (c)
Answer: a

57. Engineers who take on management roles in companies:
a) Do not have to make decisions related to allocating human resources
b) Have to ensure engineering titles are used properly
c) Are unwise to override the judgement of engineers on technical matters without investigating carefully
d) (b) and (c)
e) (a) and (b)
Answer: d

58. Which of the following factors should be taken into account in developing an action plan when faced with an ethical dilemma?
a) The effects on only some of the stakeholders affected by the dilemma
b) The circumstances surrounding the dilemma and relating to those involved
c) The details of the professional engineering code of ethics of relevance in regions unrelated to the dilemma
d) (b) and (c)

e) All of the above
Answer: b

59. Ethical dilemmas relating to engineering decisions in biomedical laboratories can arise in the area of:
a) Safeguarding the intellectual property of those who develop a new treatment technology
b) Safeguarding the rights of human subjects involved in experimental studies
c) Safeguarding the rights of people with medical devices
d) (a) and (b)
e) All of the above
Answer: d

60. The following conduct by an engineer would almost certainly be viewed as unethical by a tribunal that makes decisions on the ethical correctness of actions taken by engineers:
a) Carefully balancing the manner in which company financial resources are allocated
b) Taking appropriate action on an important risk to public safety
c) Theft of company supplies that are often used by the company, provided they are not too expensive
d) (a) and (c)
e) All of the above
Answer: c

61. The code of ethics of Professional Engineers Ontario and the IEEE code of ethics:
a) Have differences related to the different mandates of PEO and IEEE
b) Are equally important
c) Are approved by a common committee for engineering ethics
d) Both relate to licensing
e) (a), (b) and (d)
Answer: a

62. Professional engineers who establish their own company:
a) Often spend as much time dealing with decisions involving material and human resources, as well as relations between people
b) Often experience high levels of stress due to their responsibilities
c) Are not permitted to deal with technical engineering details, since they are company owners
d) (a) and (b)
e) (a) and (c)
Answer: d

63. Different parts of Ontario's code of ethics for engineers imply that engineers should:
a) Not point out technical errors made by themselves
b) Exaggerate technical errors made by another engineer
c) Point out technical errors made by another engineer
d) Not point out technical errors made by another engineer

e) All of the above
Answer: c

64. According to Ontario's code of ethics for engineers, an engineer is permitted to:
a) Explain technical issues
b) Point out risks to the welfare of the public
c) Question another engineer's competence
d) Point out technical errors of another engineer
e) All of the above
Answer: e

65. Ethical behavior is timeless and universal.
a) True
b) False
Answer: a

66. If it is legal it is always ethical.
a) True
b) False
Answer: b

67. Giving a bribe is ethical if it is part of the culture of the country.
a) True
b) False
Answer: b

68. If the impact of your decision causes harm you were probably unethical.
a) True
b) False
Answer: a

69. Organizations should be held accountable for the behavior of their suppliers.
a) True
b) False
Answer: a

70. The private sector should accept more responsibility for what was the responsibility of governments because globalization is reducing the power of governments and increasing the power of the private sector.
a) True
b) False
Answer: a

71. Being socially responsible is about making money and giving some away.
a) True

b) False
Answer: b

72. A Canadian engineer responsible for the environmental impact of his or her organization in a developing country should apply the same environmental standards as those used in Canada.
a) True
b) False
Answer: a

73. A sustainability report provides the framework for reporting on an organization's economic, environmental, and social performance.
a) True
b) False
Answer: a

74. You can never be accountable for harm caused by your decisions if you did not know your decisions would cause harm.
a) True
b) False
Answer: b

75. Ethical behavior is behavior that is in accordance with the norms of behavior of the country you are living in.
a) True
b) False
Answer: b

76. Ethical behavior includes
a) Acts, omissions, risks and dependencies for which an individual or organization is responsible to those with legitimate interest
b) The direct impact only, of decisions and activities on stakeholders
c) Respecting the rights of your employees for equitable treatment unless forbidden by law
d) None of the above
Answer: a

77. Being socially responsible is about accepting responsibility for those impacts of our decisions and activities on society and the environment, through transparent and ethical behavior
a) True
b) False
Answer: a

78. An organization is socially responsible when
a) It accepts responsibilities for the impacts of its decisions on society and the environment through ethical and transparent behavior

b) It contributes to sustainable development and the welfare of society
c) It takes into account the expectations of the stakeholders
d) It is in compliance with laws and is consistent with international norms of behavior
e) All of the above
Answer: e

79. Being socially responsible will
a) Benefit society as well as the socially responsible organization
b) Enhance competitiveness if applied strategically
c) Always reduce the risk
d) Show that "doing good" and "doing what is good for business" can be the same thing
e) All of the above
Answer: e

80. Canada's ecological footprint, per capita, is much better than the world average
a) True
b) False
Answer: b

81. The World Wildlife Fund's "Living Planet Report"
a) Confirms that we are using the planet's resources faster than they can be renewed
b) Shows that our impact upon the planet has more than tripled since 1961
c) Indicates that our footprint exceeded the world's ability to regenerate by about 25 percent
d) All of the above
Answer: d

82. Which of the following statements are true
a) Canada's GHG emissions are higher than our Kyoto commitment to be 6% below 1998 levels
b) It is not ethical to give or accept bribes even if it is the only way you can continue to operate in a country
c) Stakeholder engagement is good for an organization
d) An engineer should know what the rights and risks of harm to all of his or her stakeholders are
e) All of the above
Answer: e

83. Paying a bribe can be ethical
a) True
b) False
Answer: b

84. The degree of an organization's accountability is related to the extent of its authority
a) True

b) False
Answer: a

85. Engaging the stakeholders in an ethical and transparent way enhances accountability to the stakeholders and overall business performance
a) True
b) False
Answer: a

86. An ethical organization will ensure that the conditions of work
a) Comply with national laws and regulations where applicable
b) Are consistent with applicable international labour standards regarding wages, hours of work, weekly rest, holidays, health and safety, maternity protection, and the ability to combine work with family responsibilities
c) Allow observance of national or religious traditions and customs
d) All of the above
Answer: d

87. An organization has the responsibility to respect human rights, including within its sphere of influence
a) True
b) False
Answer: a

88. When deployed strategically, CSR can enhance innovation, productivity and competitiveness and attract investors
a) True
b) False
Answer: a

89. Socially responsible organizations do not have to give special consideration to vulnerable groups because they are the responsibility of governments
a) True
b) False
Answer: b

90. An organization should identify issues related to the impact of its activities and decisions, and the way to address these issues so as to contribute to sustainable development
a) True
b) False
Answer: a

91. Which of the following statements are true
a) An engineer should respect international norms of behavior
b) Where national law significantly conflicts with international norms, he or she should strive to

respect the norms
c) Where national law conflicts with international norms and not following these norms would have significant consequences, the engineer should review the nature of activities within that jurisdiction
d) The engineer should seek to influence relevant organizations and authorities to remedy such conflicts
e) All of the above
Answer: e

92. Canadian laws concerning workers' rights and safety ensure that workers' civil, political, social, economic, and cultural rights are met
a) True
b) False
Answer: a

93. Ethical engineers always consider the impact that their designs have on the environment and employees
a) True
b) False
Answer: a

94. Violations of human rights are increasing because of internationalization of armed conflicts and civil wars, terrorism, HIV/AIDS and other infectious diseases, environmental issues, and natural disasters
a) True
b) False
Answer: a

95. Ethical organizations identify and manage risks
a) True
b) False
Answer: a

96. Engaging with your stakeholders in an ethical and transparent way enhances accountability to stakeholders and overall business performance
a) True
b) False
Answer: a

97. An organization's responsibility for ethical labour practices only applies to the relationship of the organization with its direct employees
a) True
b) False
Answer: b

98. Organizations should ensure that the principles of social responsibility addressed in their mission, vision, operating principles, and strategy are integrated into their core processes and decision-making
a) True
b) False
Answer: a

99. The principles of social responsibility defined in ISO 26000 include
a) Recognition of stakeholders and concerns
b) Transparency
c) Respect for fundamental human rights
d) Respect for diversity
e) All of the above
Answer: e

100. Organizations need a risk management system because
a) Competent employees take shortcuts and unknowingly make major mistakes
b) All organizations have one
c) To eliminate risks
d) It helps you find the guilty party when something goes wrong
e) All of the above
Answer: a

101. Which of the following statements about sustainable development are true
a) Sustainable development is about living off the capital
b) Sustainable development is about economic growth that maximizes shareholder value
c) Sustainable development is about meeting the needs of the present generation without compromising the ability of future generations to meet their own needs
d) Sustainable development is about economic growth at the expense of the ecological system
e) Sustainable development is about balance between economic growth and the depletion of irreplaceable resources
Answer: c

102. ISO 26000 is being written to
a) Provide practical guidance related to identifying and engaging with stakeholders
b) Increase confidence and satisfaction in organizations among their customers and other stakeholders
c) Be consistent with and not in conflict with existing documents, international treaties, conventions, and ISO standards
d) All of the above
Answer: d

103. Thanks to rapid advances in engineering and technology there has been a decline in
a) The negative impact of business and society on the environment
b) Extreme poverty arising from the acceleration of globalization

c) Human rights violations by individuals, families, corporations, and civil society organizations
d) None of the above
Answer: d